

torch.linalg.tensorinv#

<https://pytorch.org/docs/stable/generated/torch.linalg.tensorinv.html>

Description:

Computes the multiplicative inverse of `torch.tensordot()`. If `m` is the product of the first `ind` dimensions of `A` and `n` is the product of the rest of the dimensions, this function expects `m` and `n` to be equal. If this is the case, it computes a tensor `X` such that `tensordot(A, X, ind)` is the identity matrix in dimension `m`. `X` will have the shape of `A` but with the first `ind` dimensions pushed back to the end. Supports input of `float`, `double`, `cfloat` and `cdouble` dtypes. When `A` is a 2-dimensional tensor and `ind=1`, this function computes the (multiplicative) inverse of `A` (see `torch.linalg.inv()`).

Function Signature

```
torch.linalg.tensorinv(A, ind=2, *, out=None) → Tensor#
```

Parameters

Keyword Arguments

Raises

Parameters

Keyword

out (Tensor, optional) – output tensor. Ignored if None. Default: None.

Code Examples

```
X.shape == A.shape[ind:] + A.shape[:ind]
```

```
linalg.tensorsolve(A, B) == torch.tensordot(linalg.tensorinv(A),  
B) # When B is a tensor with shape A.shape[:B.ndim]
```

```
>>> A = torch.eye(4 * 6).reshape((4, 6, 8, 3))  
>>> Ainv = torch.linalg.tensorinv(A, ind=2)  
>>> Ainv.shape  
torch.Size([8, 3, 4, 6])  
>>> B = torch.randn(4, 6)  
>>> torch.allclose(torch.tensordot(Ainv, B),  
torch.linalg.tensorsolve(A, B))  
True  
  
>>> A = torch.randn(4, 4)  
>>> Atensorinv = torch.linalg.tensorinv(A, ind=1)  
>>> Ainv = torch.linalg.inv(A)  
>>> torch.allclose(Atensorinv, Ainv)  
True
```

Notes & Warnings

NOTE

Note When A is a 2-dimensional tensor and `ind= 1`, this function computes the (multiplicative) inverse of A (see `torch.linalg.inv()`).

NOTE

Note Consider using `torch.linalg.tensorsolve()` if possible for multiplying a tensor on the left by the tensor inverse, as: `linalg.tensorsolve(A, B) == torch.tensordot(linalg.tensorinv(A), B)` # When B is a tensor with shape `A.shape[:B.ndim]` It is always preferred to use `tensorsolve()` when possible, as it is faster and more numerically stable than computing the pseudoinverse explicitly.

NOTE

See also `torch.linalg.tensorsolve()` computes `torch.tensordot(tensorinv(A), B)`.

Detailed Documentation

Documentation

Computes the multiplicative inverse of `torch.tensordot()`.

If `m` is the product of the first `ind` dimensions of A and `n` is the product of the rest of the dimensions, this function expects `m` and `n` to be equal. If this is the case, it computes a

tensor X such that `torch.tensordot(A, X, ind)` is the identity matrix in dimension m . X will have the shape of A but with the first `ind` dimensions pushed back to the end

Supports input of float, double, cfloat and cdouble dtypes.

When A is a 2-dimensional tensor and `ind=1`, this function computes the (multiplicative) inverse of A (see `torch.linalg.inv()`).

Consider using `torch.linalg.tensorinv()` if possible for multiplying a tensor on the left by the tensor inverse, as:

It is always preferred to use `torch.linalg.tensorinv()` when possible, as it is faster and more numerically stable than computing the pseudoinverse explicitly.

`torch.linalg.tensorinv()` computes `torch.tensordot(torch.tensorinv(A), B)`.

A (Tensor) – tensor to invert. Its shape must satisfy `prod(A.shape[:ind]) == prod(A.shape[ind:])`.

`ind` (int) – index at which to compute the inverse of `torch.tensordot()`. Default: 2.

`out` (Tensor, optional) – output tensor. Ignored if None. Default: None.

`RuntimeError` – if the reshaped A is not invertible or the product of the first `ind` dimensions is not equal to the product of the rest.